# Xiaoyang Cao

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## EDUCATION

**Massachusetts Institute of Technology** — Cambridge, MA
*M.S. in Computational Science & Engineering* — Sep 2025 – May 2027 (expected)
*Concurrent M.S. in Technology & Policy*
*Laboratory for Information and Decision Systems (LIDS)*
*Research advisor: Prof. Michiel Bakker*

**Tsinghua University** — Beijing, China
*B.S. in Mathematics and Physics* — Sep 2021 – Jun 2025

**University of California, Berkeley** — Berkeley, CA
*Exchange Student* — Jan 2024 – Aug 2024

## RESEARCH INTERESTS

How intelligence emerges in collectives of interacting agents (multi-agent reinforcement learning, populations of LLM agents) and in self-referential learning (self-play, self-distillation, preference optimization).

## PUBLICATIONS

[1] **Xiaoyang Cao**, Jingqi Li, Zhe Fu, Alexandre M. Bayen. "Who Bears the Burden? Learning Responsibility for Shared Constraints in Multi-Agent Reinforcement Learning." *Under review*, 2026.

[2] **Xiaoyang Cao**, Siddarth Srinivasan, Michiel A. Bakker. "Do Modules Stay in Their Lane? Role Drift in Compound LLM Systems." *Under review*, 2026.

[3] **Xiaoyang Cao**, Zelai Xu, Mo Guang, Kaiwen Long, Michiel Bakker, Yu Wang, Chao Yu. "RE-PO: Robust Enhanced Policy Optimization as a General Framework for LLM Alignment." *International Conference on Learning Representations (ICLR)*, 2026.

[4] **Xiaoyang Cao**, Zhe Fu, Alexandre M. Bayen. "Pareto Control Barrier Function for Inner Safe Set Maximization Under Input Constraints." *American Control Conference (ACC)*, 2025.

[5] **Xiaoyang Cao**, Dingyi Zhuang, Shenhao Wang, Jinhua Zhao. "Virtual Nodes Improve Long-term Traffic Prediction." *Transportation Research Board Annual Meeting (TRB)*, 2025.

## RESEARCH EXPERIENCE

**Learning Responsibility for Shared Constraints in MARL [1]** [**arXiv**] [**Project**] — May 2025 – Sep 2026
*Advised by Prof. Alexandre Bayen, UC Berkeley*

- Proposed LiRA (Lagrangian Responsibility Allocation): a common multiplier enforces a shared cost budget, while learned per-agent responsibility shares redistribute its penalty by optimizing social welfare, without modifying rewards or constraints.
- Proved that for convex games varying the shares induces a smooth family of normalized generalized Nash equilibria in which active constraints stay at budget while welfare varies, and derived a welfare gradient that accounts for both learning updates and the induced shift in data distribution.
- Improved average social welfare by up to 29% over uniform and agent-specific multiplier baselines across CityLearn, MABIM, Harvest, and MetaDrive (3 to 400 agents).

**Role Drift in Compound LLM Systems [2]** [**arXiv**] [**VentureBeat**] — Apr 2026 – Sep 2026
*Advised by Prof. Michiel Bakker, MIT*

- Identified *role drift*: end-to-end reinforcement learning can improve terminal performance while causing specialized modules to abandon their intended functions.
- Developed role-level evaluations showing that 86% of an apparent 31-percentage-point accuracy gain disappeared once the decomposer's assigned function was measured, demonstrating that end-to-end metrics can conceal internal failure.

- Proposed Role Anchor, a training objective that preserves the influence of role specifications while allowing modules to improve under end-to-end optimization.

**RE-PO: Robust LLM Preference Alignment** (ICLR 2026) **[3]** [**arXiv**] [**Project**] Mar 2025 – Sep 2025
*Advised by Prof. Yu Wang, Tsinghua University*

- Proposed an EM-based framework that models preference-label correctness as a latent variable, jointly estimates label and annotator reliability, and systematically robustifies DPO, IPO, SimPO, and CPO.
- Proved convergence to the true noise level under calibration and improved AlpacaEval 2 win rates by up to 7.0 percentage points on Mistral-7B and Llama-3-8B, with additional gains on Arena-Hard and MultiPref.

**Safe Control via Pareto Control Barrier Functions** (ACC 2025) **[4]** [**arXiv**] May 2024 – Oct 2024
*Advised by Prof. Alexandre Bayen, UC Berkeley*

- Developed the Pareto Control Barrier Function algorithm by integrating Pareto multi-task learning into neural control barrier functions, maximizing the inner safe set while guaranteeing safety under input constraints.
- Matched the Hamilton–Jacobi viability kernel on a 2D inverted pendulum and achieved a 700% larger inner safe set than the benchmark on a 12-D quadrotor obstacle-avoidance task.

**Virtual Nodes for Long-term Traffic Prediction** (TRB 2025) **[5]** [**arXiv**] May 2024 – Sep 2024
*Advised by Prof. Jinhua Zhao, MIT*

- Introduced virtual nodes into spatio-temporal graph neural networks to mitigate over-squashing in long-horizon traffic forecasting, using a semi-adaptive adjacency matrix learned from node embeddings.
- Reduced RMSE by 6.27% and MAPE by 5.04% on a five-year, 716-sensor San Diego dataset; the learned adjacency emphasized traffic-intensive intersections, improving interpretability.

## AWARDS & HONORS

**Outstanding Winner & INFORMS Award**, Mathematical Contest in Modeling – top 1% May 2023
**Outstanding Innovation in Science & Technology Scholarship**, Tsinghua University 2023, 2024
**Comprehensive Excellence Scholarship**, Tsinghua University Oct 2023
**Outstanding Academic Scholarship**, Tsinghua University Oct 2022

## SERVICE & TEACHING

**Conference Reviewer:** ICLR 2027, CoRL 2026
**Teaching Assistant:** Calculus A, Tsinghua University Spring 2025

## SKILLS & LANGUAGES

**ML & RL:** Python, PyTorch, Hugging Face Transformers / TRL, vLLM, DeepSpeed, FSDP, PettingZoo, RLlib
**Compute:** Multi-GPU, multi-node training on H100 / H200 / B200 clusters
**Languages:** English (fluent), Chinese (native)